Slide 1: Session opening

Main point: Decide what a test establishes and what it does not.

A benchmark score tells us how a system behaved on selected cases. It does not automatically explain why it behaved that way or whether it will work elsewhere. This session separates correctness, evidence support, sensitivity and internal mechanism. We start by specifying the task, then design controlled comparisons, then use traces and graphs to identify a repair. Each claim needs an observation that could actually support it.

Example to use A correct number with a valid citation may still have been guessed. Changing the relevant evidence is a different test from checking the citation.

Assumptions and limits Results apply to the tested task, cases, model revision and intervention. A graph is not automatically a causal explanation of the transformer.

Ask the room What do you mean when you say your model has been validated?

Slide 2: The decision this lecture supports

Main point: Distinguish scoring, diagnosis and evidence tracing.

We need to decide whether the system is ready for a stated task and which component to repair if it is not. A benchmark estimates observed performance. A diagnostic test isolates a capability. An evidence trace records the sources and operations linked to an answer. These tools complement one another but answer different questions. We will not treat one overall percentage as all three.

Example to use An extraction score can reveal a wrong date; a trace shows which passage was read; a controlled replacement checks whether that passage affected the output.

Assumptions and limits Expected behavior must be defined independently. Diagnostic conclusions depend on what was held fixed and changed.

Ask the room Which question cannot be answered by an accuracy score alone?

Slide 3: Roadmap: from a claim to a defensible test

Main point: Explain the order from task definition to release decision.

Define the task before choosing a metric. Measure separate properties, then change one relevant factor at a time. Diagnose dependencies after observing a failure and finally state the decision and limits. This sequence prevents us discovering a favorable number and inventing its meaning afterward. Keep a small record of the contract, cases, controls, settings and observations so another person can repeat the comparison.

Example to use Before testing order sensitivity, specify which records are meant to be an unordered set and which output changes would be unacceptable.

Assumptions and limits A test design can be reproducible yet poorly targeted. Reproducibility and validity both need attention.

Ask the room Which part of this sequence is usually skipped under delivery pressure?

Slide 4: Five learning objectives

Main point: Turn evaluation goals into observable work.

Participants should be able to write success and abstention conditions, choose different measures for different claims, design paired interventions, separate evidence graphs from internal causal effects and propose a repair with an owner. These are practical skills. The questions and exercise will ask you to identify the strongest conclusion a result supports rather than repeat a method name.

Example to use A useful output is a controlled test plan stating which evidence changes, what stays fixed and which observation would count as success.

Assumptions and limits The plan must fit the deployed task and its consequences; no universal metric supplies that context.

Ask the room Which objective would be most useful for an assessment you are doing now?

Slide 5: Start with an evaluation contract

Main point: Define success before measuring it.

An evaluation contract says what the system receives, what it may know, what output is required and when it should abstain. It names the intended population and excluded situations. It also specifies important errors and release tolerances. Without those decisions, a model can be marked wrong for an appropriate refusal or marked right for an unsupported guess.

Example to use For a date-extraction task, require the current effective date from an eligible source and allow withheld when the controlling source is absent.

Assumptions and limits Reference labels and source eligibility need a documented review process. The contract is a chosen task specification, not an inferred universal truth.

Ask the room What should the system do when no answer is justified?

Slide 6: A benchmark is a sample of the task

Main point: Explain coverage and independent case families.

A benchmark must include ordinary, difficult and unanswerable situations. Easy examples alone reward shortcuts. Related examples should be kept in the same family so variants of one document do not leak between development and test. Report categories as well as the total, because a high average can conceal complete failure on a critical group.

Example to use A benchmark record can contain source text Meeting moved from Tuesday to Friday, question Current day?, expected answer Friday, eligible source true, and family ID meeting-change-01. A missing-source variant should have expected behavior withhold.

Assumptions and limits These are evaluation labels, not examples to train on after inspecting test results. Sampling and family dependence govern how broadly scores can be interpreted.

Ask the room Which important failure type would an easy-only benchmark miss?

Slide 7: A labelled evaluation record

Main point: Show what is labelled and keep it separate from training.

The record contains the input, independently reviewed reference, source-support judgment and family identifier. It is not a model-generated explanation of its own correctness. A missing-source variant has a different expected behavior: the system should not invent a day. Group variants from one underlying source so they do not leak between development and final evaluation. The same labels can score several controlled conditions without changing their meaning after seeing outputs.

Example to use Case E01 asks which day applies after a Tuesday-to-Friday update. The expected answer is Friday. When the source is removed and no other support exists, the expected behavior is to withhold rather than guess.

Assumptions and limits The record is illustrative. Reference and support labels require review; evaluation fixtures must not be used to tune the final tested configuration. Related variants are dependent case families, not independent records.

Ask the room Which field would change if this were a calibration dataset rather than a final evaluation set?

Slide 8: Correctness, support and stability are different

Main point: Explain why one good property does not imply the others.

Correctness asks whether the answer matches the reference. Support asks whether eligible evidence justifies it. Stability asks whether irrelevant changes alter it. Calibration asks whether declared probabilities match outcome frequencies. A system can be stable and consistently wrong, or correct by guessing without support. Keep these assessments separate so a claim cannot silently shift from one meaning of reliable to another.

Example to use Tuesday can match a reference by chance even when no day was provided. It is correct on that row but unsupported.

Assumptions and limits Reference correctness and evidence eligibility can themselves be uncertain. Record disagreements instead of forcing every row into a false certainty.

Ask the room Can you give an example that is stable but wrong?

Slide 9: Choose the measurement for the claim

Main point: Match the observable to the assertion.

Exact extraction can use a normalized exact-match rule. Arithmetic can be independently recomputed. Free-text support needs an atomic claim, a cited span and a review rubric. Tool use needs an execution trace. None of those measurements by itself proves all the others. The measurement must capture the property actually claimed, not merely an easy proxy.

Example to use A calculator log proves which inputs were used and what arithmetic ran. It does not prove the document supplying those inputs was current or legally applicable.

Assumptions and limits Normalization, parsing and rubric choices must be recorded. A forgiving parser can hide errors, while a strict one can penalize harmless wording.

Ask the room What does your favorite metric leave unmeasured?

Slide 10: Evaluate the evaluator

Main point: Treat grading as another component that can fail.

Write examples of supported, contradicted, unsupported and unanswerable cases. Have independent reviewers apply the rubric to a shared sample and inspect disagreement. If a model assists, compare its grading with accountable expert labels and preserve reasons. A second model agreeing with the first is not independent evidence of truth by itself. Governance decisions need validated checks and ownership.

Example to use One judge may reward an answer for citing a document even when the cited paragraph says the opposite. A reviewed test case exposes that grading weakness.

Assumptions and limits Reviewer labels are a reference process with limits, not infallible truth. Inter-rater agreement is informative but does not alone prove validity.

Ask the room Where would you expect a grading model to take a shortcut?

Slide 11: Question: a correct answer cites the right document

Main point: Check the difference between support and reliance.

Ask the room to choose a conclusion. Seeing the correct answer and an apparently relevant citation is useful evidence about the output. It does not reveal which information the model actually used. It might recall the answer or reproduce it for another reason. Testing dependence requires changing the evidence under a controlled design.

Example to use A model can cite the meeting notice while answering from a memorized template. The citation alone does not tell us.

Assumptions and limits This question does not assume the model ignored the document; it says reliance has not yet been established.

Ask the room Which answer makes the narrowest justified claim?

Slide 12: Answer: B—test dependence separately

Main point: Explain the intervention needed for a stronger claim.

B is correct. First verify that the cited span supports the claim and is eligible. Then change or remove the relevant evidence and compare with a matched control. If the output moves appropriately, that supports input dependence under the tested intervention. It still does not identify a hidden neural circuit. Distinguish the level of the conclusion from the level of the experiment.

Example to use Replace Tuesday with Friday while keeping the question and irrelevant wording fixed. Check whether the answer follows the replacement.

Assumptions and limits The changed evidence must remain a coherent task input. Unnatural deletions can create distribution shifts unrelated to the claimed reasoning.

Ask the room What irrelevant edit would make a useful control?

Slide 13: Three kinds of behavioral test

Main point: Explain functionality, invariance and directional expectations.

Minimum functionality asks whether the simplest necessary capability works. Invariance asks whether a change that should not matter leaves the result stable. Directional tests ask whether relevant changes move the result in an expected direction. These are complementary. A system that is invariant to every edit could simply be ignoring all evidence.

Example to use Extract an explicit day; reorder independent records without changing content; then change the effective day and expect the answer to change.

Assumptions and limits Invariance requires a defensible claim that the edit preserves meaning and task relevance. Some record orders genuinely carry time or priority information and should not be randomized as sets.

Ask the room Which edits in your workflow should matter, and which should not?

Slide 14: Control one difference at a time

Main point: Separate intervention effects from repeat variability.

Repeat the identical prompt and evidence order to observe baseline execution variation. Then change presentation while holding facts fixed, or change one relevant fact while keeping the template fixed. Add an unrelated edit as a negative control. This prevents ordinary provider variability from being mistaken for an order or framing effect. Record settings and execution details.

Example to use Run the same evidence five times, then compare five runs after only the response-option order changes. Do not compare one baseline call with one altered call and attribute everything to order.

Assumptions and limits Temperature zero does not guarantee identical hosted execution. Paired cases and repeated controls are needed; repeats are not new independent source cases.

Ask the room What is the exact difference between your treatment and control prompts?

Slide 15: Measure changes, not just final labels

Main point: Explain action flips and probability movement separately.

Flip rate counts how often the selected action changes between paired conditions. Total variation sums the absolute changes in normalized action probabilities and divides by two. A top action can stay the same while the distribution moves substantially. Report correctness alongside both metrics: stable but wrong outputs are not a success.

Example to use A distribution moving from 0.8, 0.1, 0.1 to 0.6, 0.2, 0.2 keeps the first action on top but has total variation 0.2.

Assumptions and limits The paired action space must match and q must be a complete normalized distribution under the declared candidate-scoring method. These are language-choice probabilities, not necessarily calibrated outcome forecasts. Correlated permutations must not be counted as independent cases.

Ask the room Would a zero flip rate prove the candidate probabilities were unchanged?

Slide 16: Candidate scoring requires a fixed boundary

Main point: Explain teacher-forced sequence scores and their limits.

Score each candidate by supplying its preceding candidate tokens and adding conditional token log probabilities. All candidates must start at the same answer boundary under the same prompt. Normalize the resulting sequence scores only over the declared candidate set. This provides a conditional ranking among those strings. It is not automatically the probability an action is economically correct. Tokenization and termination can affect the comparison.

Example to use Score the literal strings BUY, CASH and WITHHOLD from the same prefix. A multi-token candidate requires all its conditional token scores, not only the first token.

Assumptions and limits Declare whether end-of-sequence is included, whether scores are length-normalized and whether candidate strings overlap. Missing top-k alternatives are unknown rather than zero. Complete candidate scoring need not agree with greedy generation.

Ask the room What would make a first-token-only comparison misleading?

Slide 17: Calibration needs outcomes and a separate split

Main point: Translate calibration into frequencies for a defined event.

Choose a specific event, such as whether an extracted field is correct. Group cases with similar predicted probabilities and compare their observed event frequency. A calibrator is fitted on one suitable labelled set, then evaluated on untouched cases. The same outcomes cannot both fit and certify the mapping. Reassess it after material changes in inputs or model behavior.

Example to use Among many cases assigned around eighty percent probability of correct extraction, roughly eighty percent should be correct under the intended population.

Assumptions and limits Labels must be valid, probabilities must refer to the same event and cases must be sufficiently representative. Small groups have wide uncertainty; drift and dependence affect interpretation.

Ask the room What exactly does the event Y equals one mean in your system?

Slide 18: Statistical evidence must match the test design

Main point: Explain the unit of analysis before naming a test.

Compare both systems on the same cases. McNemar focuses on cases where binary correctness differs and asks whether the direction of those differences is balanced. A paired bootstrap resamples independent case units while retaining the matched observations. If each source has many permutations, summarize or resample at the source-family level. Counting every permutation as an independent case makes uncertainty look too small.

Example to use Ten documents each with six orders are ten source families, not automatically sixty independent pieces of evidence.

Assumptions and limits McNemar needs independent paired case units and a suitable exact or asymptotic calculation. Bootstrap validity needs representative resampling units and enough independent groups. Multiple comparisons require a declared handling strategy.

Ask the room What is the independent unit you would resample?

Slide 19: A small p-value does not establish safe use

Main point: Explain practical tolerances and the limits of null-hypothesis tests.

A p-value concerns compatibility with a specified statistical null, not the probability that a system is safe. Report the effect size and uncertainty interval. Decide what amount of sensitivity is practically acceptable before looking at results. Failure to detect a difference does not prove equivalence; a small study may simply be uninformative. Consider how many prompts and metrics were searched.

Example to use A measured action-flip difference can be statistically detectable but tiny, while a rare severe failure can be operationally unacceptable despite wide statistical uncertainty.

Assumptions and limits Tests depend on their null, sampling unit and assumptions. Equivalence needs a prespecified tolerance and an appropriate interval or test, not merely p greater than 0.05.

Ask the room What difference would be large enough to change your release decision?

Slide 20: Class Exercise: specify a controlled evaluation

Main point: Make participants write an intervention and its controls.

Choose one claim that should follow evidence. Write a meaning-preserving edit, a relevant contradiction and an unrelated deletion. Add identical-prompt repeats. Define the expected behavior and the measurements before running anything. The purpose is to construct an interpretable comparison, not to collect a large table of outputs without a question.

Example to use Use a meeting-day task: paraphrase the notice, change Tuesday to Friday, remove an unrelated address and repeat the original prompt unchanged.

Assumptions and limits Expected effects must be independently justified. Do not label every edit invariance simply because it is small. Provider settings and score availability must be recorded.

Ask the room Ask each group which result would falsify its claim.

Slide 21: Why inspect dependencies after scoring?

Main point: Explain what an evidence trace adds to a score.

A failure rate says how often something went wrong. A dependency trace records which source span, tool inputs and derived claims were involved. That helps us choose a repair. It is especially useful when several claims share one source: a version change can affect more than the sentence where it first appears. The trace remains an external record of dependencies, not a direct view of all neural reasoning.

Example to use A wrong leverage claim may trace to a correct calculator fed the wrong cash value. The calculation itself need not be retrained.

Assumptions and limits Trace completeness and tool bindings must be verified. Missing or fabricated provenance can make a neat graph misleading.

Ask the room Which recorded handoff would distinguish a bad input from bad arithmetic?

Slide 22: An evidence graph makes the repair path visible

Main point: Read each node and edge as an explicit dependency.

Start at the source span. The tool result depends on selected values from it; a claim depends on the calculation; an action depends on the claim and the applicable rule. If the source is revised, follow outgoing dependencies to decide what needs checking again. Unrelated branches should not be invalidated without reason. The graph makes the repair scope inspectable.

Example to use Correct an eligible-cash value, recompute leverage, reconsider the headroom claim and then re-evaluate the proposed brief release.

Assumptions and limits The dependency graph must represent the relevant actual bindings. A directed edge does not prove causal mediation inside the transformer, and omitted dependencies can make impact analysis incomplete.

Ask the room Which downstream object would you recheck first after a source correction?

Slide 23: Support, dependence and mechanism form a ladder

Main point: Keep four increasingly strong claims separate.

A reviewed passage can support an answer. A controlled edit can show behavioral dependence on that passage. An internal activation intervention can test a narrower mechanism. Independent outcome evaluation can test whether a repair helps in use. Each step requires extra evidence. A citation does not jump directly to a neural mechanism or an economic benefit.

Example to use Changing a date and observing a changed answer supports input dependence. It does not by itself show which attention head carried the date.

Assumptions and limits Interventions and scopes must be specified. Internal experiments can be off distribution, and deployment claims require independent evaluation.

Ask the room Which rung does your current evidence actually reach?

Slide 24: Remove evidence; then retain only that evidence

Main point: Explain comprehensiveness and sufficiency scores without overclaiming.

Choose a target output y and score it with the full input. Remove rationale R and compare the score; a large drop suggests R mattered under that intervention. Then retain only R with the required question context and compare again; a small drop suggests R preserves the score. These are operational sensitivity measures, not proofs that the text is a complete logical argument. Redundant evidence can make removal effects small.

Example to use Full score 0.8, removal score 0.3 gives comprehensiveness 0.5. Rationale-only score 0.75 gives a sufficiency loss of 0.05. These are illustrative model scores.

Assumptions and limits Hold target, scoring boundary and necessary task instructions fixed. Declare whether p is a class or candidate score. Removal can create unnatural inputs; matched unrelated removals and coherent edits are controls.

Ask the room Could two redundant supporting passages make either one look unimportant when removed alone?

Slide 25: Patching asks a narrower internal question

Main point: Explain an activation intervention in ordinary terms.

Run a clean input and save an internal tensor. Run a related corrupted input and replace a selected internal tensor with the saved one. Continue computation and measure whether the chosen output score is restored. This tests whether that internal state can influence the outcome in the specified setting. It does not retrain the model, and it does not prove that the same mechanism always explains behavior.

Example to use Use one version with an effective date Friday and another with Tuesday, then test a declared token position and layer while keeping the remaining intervention protocol fixed.

Assumptions and limits Model weights, tensor shape, token alignment, masks and patch scope must match. Include same-run and unrelated-position controls. Artificial tensor mixtures can be off distribution; interactions can prevent additive attribution.

Ask the room How is replacing an activation different from fitting a LoRA adapter?

Slide 26: Use the graph to select tests, then verify them

Main point: Connect external dependency diagnosis to controlled model experiments.

Begin with the failed claim and trace its source spans and tool inputs. Test a source-level correction before modifying hidden states. If an internal hypothesis remains useful, map the relevant text to token positions and define a patching experiment. The graph helps choose a scope; it does not prove the mapping from a source node to a neural component. Retest any proposed repair on independent cases.

Example to use If the wrong annex was retrieved, first supply the right annex. There is little reason to patch attention before checking whether the evidence repair solves the task.

Assumptions and limits Graph-to-token mapping and causal scope are hypotheses requiring controls. A successful patch on one pair is not proof of a deployable training repair.

Ask the room What simpler intervention would you try before patching?

Slide 27: Question: no answer changes across reorderings

Main point: Check whether stability is being overstated.

The selected output stayed the same in a tested collection of orders. That is an observation about a finite set of cases and a parser. It does not prove general reliability, invariant internal states or unchanged probabilities. Ask participants to identify which unmeasured claims are being added by the stronger options.

Example to use BUY can remain the top response even while its conditional candidate score falls from eighty percent to sixty percent.

Assumptions and limits The order edits must preserve meaning and all other settings must be controlled. The test must include correctness and repeated baseline behavior.

Ask the room Which answer states only what was actually observed?

Slide 28: Answer: B—state the tested boundary

Main point: Explain how to report a negative sensitivity result honestly.

B is the warranted conclusion. Report how many cases, orders and identical repeats were tested and how answers were parsed. If full candidate scores exist, report their movement separately. Then report correctness. An unchanged wrong answer is a failure of the task, even if it is insensitive to presentation. Broader reliability requires a representative evaluation and a practical tolerance.

Example to use A result can read: no top-action flips in these tested families, with a specified maximum score shift and an independent correctness count.

Assumptions and limits Zero observed flips is not a zero population flip probability. The uncertainty depends on independent case count and test coverage.

Ask the room What additional experiment would most improve confidence in the result?

Slide 29: Recorded pilot: low sensitivity can coexist with errors

Main point: Interpret the recorded framing result without inventing a stronger story.

The expanded pilot recorded 1200 assessments across two models. It did not show the proposed directional effect of bullish framing increasing unsupported buys. Qwen still mishandled some evidence gaps, while Llama passed the specified benchmark. These observations demonstrate why sensitivity and correctness must be reported separately. They do not identify why a subjective statement was ignored internally.

Example to use A model can give the same unsupported recommendation under neutral, supportive and opposing framing. The lack of a framing effect does not rescue the missing-evidence error.

Assumptions and limits The counts and model findings refer to the recorded synthetic pilot. They are not prevalence estimates for all prompts, models or financial decisions. No new internal-mechanism experiment is implied.

Ask the room Which finding is about framing, and which is about evidence sufficiency?

Slide 30: What BeliefLens adds to this evaluation loop

Main point: Describe concrete outputs without attributing untested powers.

BeliefLens organizes source records, executed operations and derived claims into inspectable dependencies. Declared requirements can flag unresolved support or contradictions and identify affected claims. Controlled-run comparisons can then be linked back to the changed evidence. Repairs should carry a record of what changed and which tests were repeated. This supports diagnosis and review rather than replacing domain judgment.

Example to use A missing source node can mark a downstream headroom claim unresolved and route the issue to the responsible reviewer.

Assumptions and limits The available inputs, graph construction and rule definitions bound what can be detected. Unknown sources or hidden selection bias are not automatically discovered. Separate implemented checks from proposed algorithms.

Ask the room What concrete record would let an owner decide whether a repair is complete?

Slide 31: Finish with an evaluation card

Main point: Leave an inspectable account of the claim and its limits.

Record the task, intended population, reference process, independent case families, controls, model version and key measurements. Include uncertainty intervals, known gaps and a named owner for retesting. A compact card makes it harder for a score to lose its context as it moves into a leadership presentation. It should allow someone to locate the evidence behind the release recommendation.

Example to use An evaluation card can state that a model passed date extraction but has not been qualified for unsourced interpretation or autonomous release.

Assumptions and limits A card summarizes evidence; it does not replace access to the underlying cases, logs and independent checks.

Ask the room Which sentence would prevent a reader from overgeneralizing your result?

Slide 32: Five objectives: what can you now demonstrate?

Main point: Check that each objective has an observable deliverable.

Ask for a task contract, a justified metric, a paired-control plan, a careful causal claim and a repair record. These are the products of the session. Do not ask participants merely to repeat names such as total variation or patching. The key skill is choosing the test whose observation answers the actual question.

Example to use A participant can explain why a citation check establishes support but not reliance, then design the relevant evidence edit.

Assumptions and limits The strength of any resulting claim remains bounded by cases, controls, label quality and assumptions.

Ask the room Ask someone to name one claim they would now narrow after this session.

Slide 33: Reading route: lectures and primary methods

Main point: Explain where to look for deeper treatment of a specific test.

Use evaluation lectures for benchmark design and metrics, CheckList for functionality and invariance tests, rationale work for removal and retention scores, and interpretability papers for controlled internal interventions. These sources should deepen a defined question. A reference to an established method is not evidence that a particular implementation or benchmark is valid.

Example to use If a team claims evidence dependence, read the intervention protocol rather than rely on a generic accuracy tutorial.

Assumptions and limits Each method has its own task, scoring and distribution assumptions. Preserve source attribution and avoid presenting an adapted illustration as a published result.

Ask the room Which source would help resolve your next methodological uncertainty?